Skills Summary

One, No, or Infinitely Many Solutions

Use this reference while completing your worksheet or when studying.

The whole method in one move: simplify each side, then get the variable terms together. What is left decides everything.

- A variable term survives ⇒ **one solution**.
- The variable cancels and the leftover statement is **false** ⇒ **no solution**.
- The variable cancels and the leftover statement is **true** ⇒ **infinitely many solutions**.

Skill 1 — Equations with exactly one solution

Test: after simplifying, the two sides have *different* numbers of the variable. Then one subtraction leaves a single variable term, and dividing gives the one value that works.

Example: Solve and classify $5x + 2 = 3x + 10$.
Step 1. The sides carry different amounts of x (five of them against three), so the variable cannot cancel — expect exactly one solution.
Step 2. Subtract $3x$: $2x + 2 = 10$. Subtract 2: $2x = 8$. Divide by 2:
$x = 4$, one solution. $\boldsymbol{x = 4}$

Try it: Solve and classify $7x + 1 = 2x + 16$.

check: $\boldsymbol{x = 3}$

Skill 2 — Equations with no solution

Test: the variable terms match exactly, but the constants do not. Subtracting the variable term leaves a false number sentence, and no value of the variable can repair it.

Example: Solve and classify $4x + 9 = 4x + 2$.
Step 1. Both sides carry the same amount of x, so the variable will cancel — the constants alone will decide the answer.
Step 2. Subtract $4x$ from both sides: $9 = 2$, and the two constants differ by 7, so the sentence is false.
No value of x works. $\boldsymbol{7 \neq 0}$, no solution

Try it: Solve and classify $6x + 5 = 6x - 1$.

check: $\boldsymbol{6 \neq 0}$, no solution

(!) Watch out: "No solution" is not the same as "$x = 0$". $x = 0$ means zero works; no solution means nothing works. Write the words.

Skill 3 — Equations true for every value

Test: after distributing, the two sides are the *same expression*. Subtracting leaves a true sentence such as $0 = 0$, so every number is a solution.

Example: Solve and classify $3(2x + 4) = 6x + 12$.
Step 1. The left side is not simplified yet, so distribute first and then compare the two sides term by term.
Step 2. Distribute: $3(2x + 4) = 6x + 12$, which is exactly the right side.
Every value of x works. $\boldsymbol{6x + 12 = 6x + 12}$

Try it: Solve and classify $5(x - 2) = 5x - 10$.

check: $\boldsymbol{5x - 10 = 5x - 10}$, infinitely many

(!) Watch out: distribute to *every* term inside the parentheses. Forgetting the second one turns an "infinitely many" equation into a fake one-solution answer.

Key Vocabulary

Solution: a value that makes the equation a true statement.

Identity: an equation true for every value of the variable.

Like terms: terms with the same variable part, which can be combined.

Distribute: multiply the outside factor by each term inside.